

① Ethics, Law and professional in Software Engineering

(a) Differentiate between ethics, law and professional in computing and explain how these three concepts may lead to different decisions in the above

Situation: provide suitable computing-related examples.

Ethics, Law and professional responsibility are three important concepts in computing but are not the same.

Ethics	Law	professional Responsibility
Principles that help us decide what is right or wrong.	Rules made by the government that every one must follow.	Duties that a Software professional should follow towards users, employers and society.
Ex: Not sharing users private data without permission.	Ex: following privacy and data protection laws.	Ex: Reporting a serious security or privacy problem to management.

In this given situation: ①

The health app collects location, heart-rate and lifestyle data. The marketing team wants to use this data for targeted advertising without getting additional consent.

Ethics: Using the data without user's permission

is ~~not~~ unethical because it violates their privacy and trust.

Law: If applicable privacy law require consent for this type of data use, using it without consent would be illegal.

Professional Responsibility: A software engineer

should protect user's privacy and safety and

should inform management about the problem, even if the company want to increase its profit.

(b) Two Historical Cases of Ethical Failure in Technology.

1. Volkswagen Emission Scandal

- Technology involved : Car engine-control software.
- Ethical failure : Volkswagen used software to make cars appear to meet pollution standards during official tests, while producing higher emissions during normal driving.
- Stakeholders affected : Customers, employees, government regulators, the public and the environment.
- Lessons learned : Software should not be used to deceive people or regulators.

Computer and engineers should give priority to honesty, safety, and environmental responsibility.

2. Boeing 737 MAX Crashes

- Technology involved: Aircraft flight-control software, including the MCAS system.
- Ethical failure: problems related to software design, testing, safety management and communication contributed to two fatal crashes.
- Stakeholders affected: passengers, pilots, their families, airlines, employees, and aviation regulators.
- Lessons learned: Safety-critical software must be properly designed, tested and monitored. Business goals should never be more important than human safety.

⌚: 7 min

(C) Major Principles of Software Engineering Ethics

→ Software engineering ethics means following moral and professional principles while developing and maintaining software. The major principles are:-

1. Public Safety: Software engineers should make sure that software does not harm people or society. Safety is especially important in medical, banking and transportation systems.

2. Privacy: Engineers should protect user's personal information and should not collect or share data without proper permission.

3. Security: Engineers should protect software and data from unauthorized access, hacking and misuse.

4. Honesty: Engineers should be honest about the software's features, limitations, risks, and problems. They should not provide false information.

5. Competence: Engineers should have the necessary knowledge and skills to do their work properly. They should also continue learning new technologies.

6. Interest of users: Engineers should consider the needs, rights and well-being of users when designing and developing software.

Professional Responsibility:

Professional responsibility means that Software engineers must consider the effects of their work on users and society, not only the company's goals. For example if a health app collects user's personal data, engineers should ensure that the data is secure, private and used responsibly.

⌚: 10min

① ACM/IEEE: Software engineering code of Ethics.

The ACM/IEEE Software Engineering code of Ethics provides guidelines that help software engineers make ethical and professional decisions.

When the company's objective conflict with the public interest developers should consider the following principle.

1. public Interest: Developers should give priority to public safety, privacy and welfare.

2. Client and Employer: Developers should support their employer, but they should not do anything that seriously harms the public.

3. Product Quality: Software should be safe, reliable, secure and properly tested.

4. Professional judgment: Developers should make honest and independent decisions and report serious problem.

5. Management: Manager and Developers should support other professional.

6. Colleagues: Developers should respect and support other professional who raise ethical concerns.

7. Continuous learning: Developers should maintain their skills and knowledge to perform their responsibilities properly.

⌚: 70 min

2. Ethical frameworks and Software privacy Dilemma.

(a) Compare and contrast Utilitarianism and Deontology. Explain how each framework would evaluate the company's proposed data-collection practice.

Utilitarianism	Deontology
(i) An action is right if it produces the greatest overall benefit. (ii) The company may collect the data if the benefits such as better recommendation and improve user experience are	(i) An action is right when it follows moral duties and respects people's rights, regardless of the result. (ii) The company should not analyze private message or personal data without proper consent because it violates user's privacy and

greater than the privacy.

autonomy

harms.

Utilitarianism :

A utilitarianism would compare the benefits and harms of collecting the data.

Benefits:

- Better recommendations.
- More useful services.
- Higher user engagement.

Harms :

- Loss of privacy.
- possible misuse of personal information.
- Loss of user trust.

Deontology:

A deontology approach focuses on rules, duties and individual rights rather than the final result.

Users have a right to privacy and control over their personal information. Therefore, it

analyzing private messages and contact networks without informed consent would be considered wrong, even if it improves the services.

⌚: 8 min

(b) Explain virtue Ethics and discuss how qualities such as honesty, fairness, responsibility, courage, and respect AI could influence the decision of a Software engineer.

Virtue Ethics is an ethical approach that focuses on the character and qualities of a person rather than only rules or consequences. It asks "What would a good and responsible person do?"

In Software engineering, important virtue can guide engineers when making difficult decisions.

1. Honesty : The engineer should be truthful about how the AI system collects and uses users' data. They should not hide privacy risks from users or management.

2. Fairness: The engineer should treat all users fairly and avoid creating a system that unfairly disadvantages certain users.

3. Responsibility: The engineer should take responsibility for the effect of the software and protect user's personal information.

4. Courage: If the company wants to collect private message without proper consent, the engineer should have the courage to raise the concern and question the decision, even if it is difficult.

5. Respect: The engineer should respect users' privacy, right, and choice. Users should have control over how their personal information

2(c) Ethical Construct and explain an ethical decision-making flow based on the following structure

Ende

Identify the ethical issue



Identify stakeholders and potential harms



Apply ethical principles / frameworks



Check legal and professional obligations



Evaluate possible alternatives



select and justify the most ethical action.

An ethical decision-making flow helps software engineers make responsible decisions when they face an ethical problem.

Explanation:

1. Identify the ethical issue:

First, identify what is ethically wrong or concerning.

Example: The social-media company wants to analyze user's private message. without proper consent.

2. Identify stakeholders and potential harms:

Identify who may be affected and what harm may occur.

Stakeholders: users, company, employees, advertisers, and society.

Potential harms: Loss of privacy, misuse of personal data, and loss of user trust.

3. Apply ethical principle / framework:

Use ethical theories such as Utilitarianism, Deontology and virtue Ethics. Consider privacy, fairness, honesty and user right.

4. Check legal and professional obligations:

Check whether the proposed action follows privacy

laws and professional codes such as the

AAC/IEEE Software Engineering code of Ethics.

5. Evaluate possible alternatives:

Consider different solutions:

- Do not collect private message.

- collect only necessary data.

- Ask users for informed consent.

- Anonymize or minimize the collected data.

6. Select and justify the most ethical actions.

⌚ : 8 min

(d) Explain why an ethical decision may still be required even when the proposed activity is technically legal.

⇒ An activity can be legal allowed but still unethical. Laws provide minimum rules for acceptable behavior, while ethics consider what is right, fair and responsible.

1. Law does not cover every situation

Some harmful actions may not be specifically prohibited by law.

2. User trust matters. Companies should respect user expectations and privacy.

3. Ethics considers fairness and harm.

4. Professional responsibility goes beyond

legal compliance.

5. Technology change quickly.

Example: A company may legally collect

browsing data, but using that data in a way that

user did not reasonably expect may still be

unethical.

3. Explainability and Black-Box AI in High-Stakes Decision.

(a) Explain why transparency and explainability

are important ethical requirements for AI System. used in high-stakes domains such as credit lending, healthcare, employment and criminal justice.

Transparency means making it clear how an AI system works, what data it uses, and how decisions are made.

Explainability means being able to give a clear and understandable reason for a particular AI decision.

Importance:

1. Fairness: Explainability helps identify whether an AI system is making unfair or biased decisions against a particular group.
2. Accountability: Organizations should be able to explain and take responsibility for AI decisions instead of simply saying "The AI decided it".
3. User trust: People are more likely to trust an AI system when they understand why a decision was made.
4. Right to challenge: If a person is rejected for a loan, they should be able to understand the reason and where appropriate, correct incorrect

information or challenge the decision.

5. Finding Errors and Bias.

6. Legal and professional Responsibility.

(b) Discuss at least two societal harms

that may occur when an AI system operates as black box.

A black box AI system is a system where people cannot easily understand how the AI makes its decision. They can create several harms in society.

1. Discrimination and Bias.

Example: In the given loan system, if applicants from a particular community are rejected

more often without a clear reason.

2. Lack of Accountability: When a AI decision

cannot be explained, it becomes difficult to determine who is responsible when something goes wrong.

3. Loss of trust: People may lose trust in organization when important decision are made by system.

4. Difficulty in Correcting Errors:

⌚: 7min

(c) Explain how lack of explainability can affect procedural fairness, accountability, public trust and an individual's ability to challenge

Lack of explainability means people cannot clearly understand why an AI system made a particular decision.

1. Procedural fairness: people should be treated fairly during the decision making process.

2. Accountability: when the AI's decision cannot be explained.

3. public trust: people are less likely to trust an AI system when they do not understand how it makes important decisions.

4. Ability to Challenge a Decision: A person needs to know the reason for a decision in order to correct errors or challenge it.

⌚: 6 min

3(d) Discuss what information should ideally be provided to an affected person when an AI system makes an important decision about them.

When an AI system makes an important decision about a person, the affected person should receive clear and understandable information about the decision.

1. Reason for the Decision: The person should know why the AI made the decision.

2. Important Factors used: The organization should explain the main factors or type of

data that influenced the decision.

3. Use of AI: The person should be

informed that an AI or automated system

was involved in making the decision.

4. possible error: They should be told how incorrect or outdated information could affect the decision.

5. Right to Review or Appeal.

6. How to correct information.

7. Contact Information.

⌚: 7min

(4) ACM code of Ethics and Professional Decision Making

(a) Explain the key principle of the ACM code of Ethics and why professional codes are important for computing network professionals.

The ACM (Association for Computing Machinery) Code of Ethics is a set of professional guidelines

key principle of the ACM code of Ethics

1. Contribute to society and Human well-Being.

2. Avoid Harm: professionals should prevent actions that may harm users, organizations.

3. Be Honest and Trustworthy.

4. Respect privacy : Personal information should be collected.

5. Maintain Security and confidentiality.

6. Personal competence :

Importance of professional codes :

- provide guidance for making ethical decision.
- Helps professional act responsibly and

Consistently :

- protect users, organizations and society from harm.

- promote honesty, fairness, Privacy and Security.

⌚ : 7 min

(b) Identify the ACM principles that are relevant to the above situation and explain how they should influence the engineer's decision.

In this situation the manager wants to release the system even though testing has found serious security weakness.

1. Contribute Society and Human well-Being.
2. Avoid Harm.
3. Be Honest and Trustworthy.
4. Respect privacy
5. Maintain professional competence
- 6..

⌚ : 5 min

(c) Discuss the responsibilities of Software Professionals when organization pressure conflicts with public safety, security or user interests.

Software professionals have important responsibilities when organization pressure conflicts with public safety, security or user interest.

1. protect public safety.

2. protect security.

3. protect user interests.

4. Be honest.

5. Raise ethical concerns.

6. use professional judgment.

7. Refuse Harmful Action when necessary.

⌚: 7min

(d) Explain how professional code can support ethical decision making without replacing an individual's professional judgment.

⇒ How a professional code support Decision Making.

1. provides Guidance.
2. Helps Identify Ethical problem.
3. provides a common Standard.
4. Support Difficult Decision.
5. Encourage Responsibility.

Why professional judgment is still

Not Necessary.

A code cannot provide an exact answer for every situation. Different situation may involve different risks, laws, users and ethical conditions. Therefore, engineers must use their own knowledge, experience, critical thinking and understanding of the situation.

④: 7min

5. Whistleblowing , Professional Dissent and Ethical Responsibility.

(a) Define professional dissent and whistleblowing in the context of computing.

Professional Dissent :

Professional Dissent means respectfully disagreeing with a decision or practice at the workplace because it may be unsafe, unethical or harmful. A software engineer may raise concerns with a manager or team when they believe a decision could harm users.

Whistleblowing:

Whistleblowing means reporting serious wrongdoing illegal activities or dangerous practice to an appropriate authority, especially when normal internal channels do not solve the problem.

⌚: 7 min

(b) Explain the ethical considerations that the analyst should evaluate before deciding whether to disclose the issue externally.

Before deciding to disclose the issue externally, the cybersecurity analyst should carefully evaluate several ethical factors:

1. Severity of Harm: The analyst should determine how serious the vulnerability is and whether it could cause significant harm to customers or the public.

2. Public safety and security.

3. Internal Reporting effort.

4. Evidence and Accuracy.

5. Legal and professional obligations.

6. possible consequence : The analyst should use as the effects of disclosure on customers.

the company, before deciding to disclose.

7. Appropriate Disclosure Method: If external reporting because necessary the Analyst

Should use appropriate and lawful channels

to minimize unnecessary harm.

④: 7 min

6(a) Explain how the AEM principles "Contribute to Society and human well-being" and "Respect privacy" may come into conflict.

The AEM code of Ethics says computing professionals should contribute to society and human well-being and also respect privacy.

1. Contribute to society and human well-being.
2. Respect privacy.

Why they conflict:

The research may provide a large social benefit, but collecting or analyzing detailed location

Example: If researchers use location data to

Identify areas where an infectious disease is spreading, the information could help authorities respond quickly.

Ethical Balance:

- use anonymized or aggregated data.
- Collect only the data that is necessary
- Limit access to sensitive information.
- use strong security measures.
- Clearly define how the data will be used and stored.

⌚ : 8 min